

# Network switch

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A **network switch** is a computer networking device that connects devices together on a computer network, by using a form of packet switching to forward data to the destination device. A network switch is considered more advanced than a (repeater) hub because a switch will only forward a message to one or multiple devices that need to receive it, rather than broadcasting the same message out of each of its ports.<sup>[1]</sup>



Avaya ERS 2550T-PWR, a 50-port network switch

A network switch (also called **switching hub**, **bridging hub**, officially **MAC bridge**<sup>[2]</sup>) is a multi-port network bridge that processes and forwards data at the data link layer (layer 2) of the OSI model. Switches can also incorporate routing in addition to bridging; these switches are commonly known as *layer-3* or *multilayer switches*.<sup>[3]</sup> Switches exist for various types of networks including Fibre Channel, Asynchronous Transfer Mode, InfiniBand, Ethernet and others. The first Ethernet switch was introduced by Kalpana in 1990.<sup>[4]</sup>

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## Overview

A switch is a device used on a computer network to physically connect devices together. Multiple cables can be connected to a switch to enable networked devices to communicate with each other. Switches manage the flow of data across a network by only transmitting a received message to the device for which the message was intended. Each networked device connected to a switch can be identified using a MAC address, allowing the switch to regulate the flow of traffic. This maximises security and efficiency of the network.

Because of these features, a switch is often considered more "intelligent" than a network hub. Hubs neither provide security, or identification of connected devices. This means that messages have to be transmitted out of every port of the hub, greatly degrading the efficiency of the network.

## Network design

An Ethernet switch operates at the data link layer of the OSI model to create a separate collision domain for each switch port. Each computer connected to a switch port can transfer data to any of the other ones at a time, and the transmissions will not interfere with the limitation that, in half duplex mode, each line can only *either* receive from *or* transmit to its connected computer at a certain time. In full duplex mode, each line can simultaneously transmit *and* receive, regardless of the partner.

In the case of using a repeater hub, only a single transmission could take place at a time for all ports combined, so they would all share the bandwidth and run in half duplex. Necessary arbitration would also result in collisions requiring retransmissions.

## Applications

The network switch plays an integral part in most modern Ethernet local area networks (LANs). Mid-to-large sized LANs contain a number of linked managed switches. Small office/home office (SOHO) applications typically use a single switch, or an all-purpose converged device such as a residential gateway to access small office/home broadband services such as DSL or cable Internet. In most of these cases, the end-user device contains a router and components that interface to the particular physical broadband technology. User devices may also include a telephone interface for Voice over IP (VoIP) protocol.

## Microsegmentation

Segmentation is the use of a bridge or a switch (or a router) to split a larger collision domain into smaller ones in order to reduce collision probability and improve overall throughput. In the extreme,



Cisco small business  
SG300-28 28-port Gigabit  
Ethernet rackmount switch and  
its internals

i. e. microsegmentation, each device is located on a dedicated switch port. In contrast to an Ethernet hub, there is a separate collision domain on each of the switch ports. This allows computers to have dedicated bandwidth on point-to-point connections to the network and also to run in full-duplex without collisions. Full-duplex mode has only one transmitter and one receiver per 'collision domain', making collisions impossible.

## Role of switches in a network

Switches may operate at one or more layers of the OSI model, including the data link and network layers. A device that operates simultaneously at more than one of these layers is known as a *multilayer switch*.

In switches intended for commercial use, built-in or modular interfaces make it possible to connect different types of networks, including Ethernet, Fibre Channel, ATM, ITU-T G.hn and 802.11. This connectivity can be at any of the layers mentioned. While layer-2 functionality is adequate for bandwidth-shifting within one technology, interconnecting technologies such as Ethernet and token ring is easier at layer 3.

Devices that interconnect at layer 3 are traditionally called routers, so layer-3 switches can also be regarded as (relatively primitive) routers.

Where there is a need for a great deal of analysis of network performance and security, switches may be connected between WAN routers as places for analytic modules. Some vendors provide firewall, [5][6] network intrusion detection,<sup>[7]</sup> and performance analysis modules that can plug into switch ports. Some of these functions may be on combined modules.<sup>[8]</sup>

In other cases, the switch is used to create a mirror image of data that can go to an external device. Since most switch port mirroring provides only one mirrored stream, network hubs can be useful for fanning out data to several read-only analyzers, such as intrusion detection systems and packet sniffers.

## Layer-specific functionality

While switches may learn about topologies at many layers, and forward at one or more layers, they do tend to have common features. Other than for high-performance applications, modern commercial switches use primarily Ethernet interfaces.

At any layer, a modern switch may implement power over Ethernet (PoE), which avoids the need for attached devices, such as a VoIP phone or wireless access point, to have a separate power supply. Since switches can have redundant power circuits connected to uninterruptible power supplies, the connected device can continue operating even when regular office power fails.

### Layer 1 (Hubs versus higher-layer switches)

A network hub, or repeater, is a simple network device. Repeater hubs do not manage any of the traffic that comes through them. Any packet entering a port is flooded out or "repeated" on every other port, except for the port of entry. Since every packet is repeated on every other port, packet collisions affect the entire network, limiting its capacity.

A switch creates the – originally mandatory – Layer 1 end-to-end connection only virtually. Its bridge function selects which packets are forwarded to which port(s) on the basis of information taken from layer 2 (or higher), removing the requirement that every node be presented with all data. The connection lines are not "switched" literally, it only appears like this on the packet level. "Bridging hub", "switching hub", or "multiport bridge" would be more appropriate terms.

There are specialized applications where a hub can be useful, such as copying traffic to multiple network sensors. High end switches have a feature which does the same thing called port mirroring.

By the early 2000s, there was little price difference between a hub and a low-end switch.<sup>[9]</sup>

## Layer 2

A network bridge, operating at the data link layer, may interconnect a small number of devices in a home or the office. This is a trivial case of bridging, in which the bridge learns the MAC address of each connected device.

Single bridges also can provide extremely high performance in specialized applications such as storage area networks.

Classic bridges may also interconnect using a spanning tree protocol that disables links so that the resulting local area network is a tree without loops. In contrast to routers, spanning tree bridges must have topologies with only one active path between two points. The older IEEE 802.1D spanning tree protocol could be quite slow, with forwarding stopping for 30 seconds while the spanning tree reconverged. A Rapid Spanning Tree Protocol was introduced as IEEE 802.1w. The newest standard Shortest path bridging (IEEE 802.1aq) is the next logical progression and incorporates all the older Spanning Tree Protocols (IEEE 802.1D STP, IEEE 802.1w RSTP, IEEE 802.1s MSTP) that blocked traffic on all but one alternative path. IEEE 802.1aq (Shortest Path Bridging SPB) allows all paths to be active with multiple equal cost paths, provides much larger layer 2 topologies (up to 16 million compared to the 4096 VLANs limit),<sup>[10]</sup> faster convergence, and improves the use of the mesh topologies through increase bandwidth and redundancy between all devices by allowing traffic to load share across all paths of a mesh network.<sup>[11][12][13][14]</sup>

While *layer 2 switch* remains more of a marketing term than a technical term, the products that were introduced as "switches" tended to use microsegmentation and Full duplex to prevent collisions among devices connected to Ethernet. By using an internal forwarding plane much faster than any interface, they give the impression of simultaneous paths among multiple devices. 'Non-blocking' devices use a forwarding plane or equivalent method fast enough to allow full duplex traffic for each port simultaneously.

Once a bridge learns the addresses of its connected nodes, it forwards data link layer frames using a layer 2 forwarding method. There are four forwarding methods a bridge can use, of which the second



A modular network switch with three network modules (a total of 24 Ethernet and 14 Fast Ethernet ports) and one power supply.

through fourth method were performance-increasing methods when used on "switch" products with the same input and output port bandwidths:

1. Store and forward: The switch buffers and verifies each frame before forwarding it.
2. Cut through: The switch reads only up to the frame's hardware address before starting to forward it. Cut-through switches have to fall back to store and forward if the outgoing port is busy at the time the packet arrives. There is no error checking with this method.
3. Fragment free: A method that attempts to retain the benefits of both store and forward and cut through. Fragment free checks the first 64 bytes of the frame, where addressing information is stored. According to Ethernet specifications, collisions should be detected during the first 64 bytes of the frame, so frames that are in error because of a collision will not be forwarded. This way the frame will always reach its intended destination. Error checking of the actual data in the packet is left for the end device.
4. Adaptive switching: A method of automatically selecting between the other three modes.

While there are specialized applications, such as storage area networks, where the input and output interfaces are the same bandwidth, this is not always the case in general LAN applications. In LANs, a switch used for end user access typically concentrates lower bandwidth and uplinks into a higher bandwidth.

### Layer 3

Within the confines of the Ethernet physical layer, a layer-3 switch can perform some or all of the functions normally performed by a router. The most common layer-3 capability is awareness of IP multicast through IGMP snooping. With this awareness, a layer-3 switch can increase efficiency by delivering the traffic of a multicast group only to ports where the attached device has signaled that it wants to listen to that group.

### Layer 4

While the exact meaning of the term *layer-4 switch* is vendor-dependent, it almost always starts with a capability for network address translation, but then adds some type of load distribution based on TCP sessions.<sup>[15]</sup>

The device may include a stateful firewall, a VPN concentrator, or be an IPSec security gateway.

### Layer 7

Layer-7 switches may distribute loads based on **Uniform Resource Locator** URL or by some installation-specific technique to recognize application-level transactions. A layer-7 switch may include a web cache and participate in a content delivery network.<sup>[16]</sup>

## Types of switches

### Form factor

- *Desktop*, not mounted in an enclosure, typically intended to be used in a home or office environment outside of a wiring closet.
- *Rack-mounted*, a switch that mounts in an equipment rack.
- *Chassis*, with swappable module cards.
- *DIN rail-mounted*, normally seen in industrial environments.



Rack-mounted 24-port 3Com switch

## Configuration options

- *Unmanaged* switches – these switches have no configuration interface or options. They are plug and play. They are typically the least expensive switches, and therefore often used in a small office/home office environment. Unmanaged switches can be desktop or rack mounted.
- *Managed* switches – these switches have one or more methods to modify the operation of the switch. Common management methods include: a command-line interface (CLI) accessed via serial console, telnet or Secure Shell, an embedded Simple Network Management Protocol (SNMP) agent allowing management from a remote console or management station, or a web interface for management from a web browser. Examples of configuration changes that one can do from a managed switch include: enabling features such as Spanning Tree Protocol or port mirroring, setting port bandwidth, creating or modifying Virtual LANs (VLANs), etc. Two sub-classes of managed switches are marketed today:
  - *Smart* (or intelligent) switches – these are managed switches with a limited set of management features. Likewise "web-managed" switches are switches which fall into a market niche between unmanaged and managed. For a price much lower than a fully managed switch they provide a web interface (and usually no CLI access) and allow configuration of basic settings, such as VLANs, port-bandwidth and duplex.<sup>[17]</sup>
  - *Enterprise Managed* (or fully managed) switches – these have a full set of management features, including CLI, SNMP agent, and web interface. They may have additional features to manipulate configurations, such as the ability to display, modify, backup and restore configurations. Compared with smart switches, enterprise switches have more features that can be customized or optimized, and are generally more expensive than smart switches. Enterprise switches are typically found in networks with larger number of switches and connections, where centralized management is a significant savings in administrative time and effort. A stackable switch is a version of enterprise-managed switch.

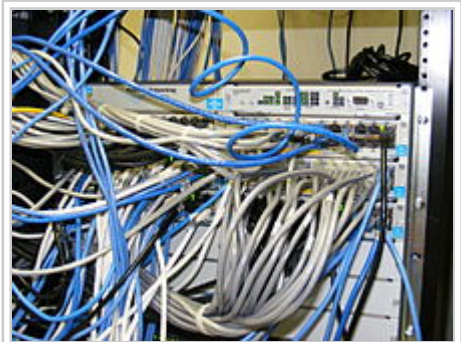
## Typical switch management features



- Turn particular port range on or off
- Link bandwidth and duplex settings
- Priority settings for ports
- IP Management by IP Clustering
- MAC filtering and other types of "port security" features which prevent MAC flooding
- Use of Spanning Tree Protocol
- SNMP monitoring of device and link health
- Port mirroring (also known as: port monitoring, spanning port, SPAN port, roving analysis port or link mode port)
- Link aggregation (also known as *bonding*, *trunking* or *teaming*) allows the use of multiple ports for the same connection achieving higher data transfer rates
- VLAN settings. Creating VLANs can serve security and performance goals by reducing the size of the broadcast domain
- 802.1X network access control
- IGMP snooping



Linksys 48-port switch



HP Procurve rack-mounted switches mounted in a standard Telco Rack 19-inch rack with network cables

## Traffic monitoring on a switched network

Unless port mirroring or other methods such as RMON, SMON or sFlow are implemented in a switch,<sup>[18]</sup> it is difficult to monitor traffic that is bridged using a switch because only the sending and receiving ports can see the traffic. These monitoring features are rarely present on consumer-grade switches.

Two popular methods that are specifically designed to allow a network analyst to monitor traffic are:

- Port mirroring – the switch sends a copy of network packets to a monitoring network connection.
- SMON – "Switch Monitoring" is described by RFC 2613 and is a protocol for controlling facilities such as port mirroring.

Another method to monitor may be to connect a layer-1 hub between the monitored device and its switch port. This will induce minor delay, but will provide multiple interfaces that can be used to monitor the individual switch port.

## See also

- Bridging (networking)
- Console server
- Energy-Efficient Ethernet
- Fibre Channel switch
- Fully switched network
- LAN switching
- Local area network
- Packet switch
- Router (computing)
- Stackable switch
- Telephone exchange
- Turing switch
- Wide area network

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## External links

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- Understanding the different types of Ethernet Switches - Cisco Blogs (<http://blogs.cisco.com/smallbusiness/understanding-the-different-types-of-ethernet-switches/>)

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